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M T W T F S S

$$1. a. W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x - 2y + z = 1 \right\}$$

it is not a subspace because $(0, 0, 0)$ does not satisfy $x - 2y + z = 1$, i.e. $(0, 0, 0) \notin W$

So, W is not a subspace

$$b. W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x^2 + y^2 - z = 0 \right\}$$

let $(1, 1, 2)$ and $(2, 1, 5) \in W$ because it satisfies $x^2 + y^2 - z = 0$

Now

$$(1, 1, 2) + (2, 1, 5) = (3, 2, 7) \notin W$$

$$\text{because } 3^2 + 2^2 - 7 = 9 + 4 - 7 = 6 \neq 0$$

$$c. W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x - 2y + z = 0 \right\}$$

clearly, $(0, 0, 0) \in W$

(x_1, y_1, z_1) and $(x_2, y_2, z_2) \in W$

$$\text{i.e. } x_1 - 2y_1 + z_1 = 0 \quad \text{and} \quad x_2 - 2y_2 + z_2 = 0 \quad \text{--- (1) and (2)}$$

Now,

$$(x_1 + x_2) - 2(y_1 + y_2) + (z_1 + z_2)$$

$$= (x_1 - 2y_1 + z_1) + (x_2 - 2y_2 + z_2)$$

$$= 0 + 0 = 0$$

$$(x_1 + x_2, y_1 + y_2, z_1 + z_2) \in W$$

Now,

$$\lambda x_1 - 2y_1 + \lambda z_1 = \lambda(x_1 - 2y_1 + z_1)$$

$$= \lambda \cdot 0 = 0$$

$$\lambda(x_1, y_1, z_1) \in W$$

By subspace test, W is a subspace of $\mathbb{R}^3$.

$$2. \quad v_1 = \begin{bmatrix} 3 \\ 2 \\ -3 \\ 4 \end{bmatrix} \quad v_2 = \begin{bmatrix} -1 \\ 3 \\ 4 \\ -12 \end{bmatrix} \quad v_3 = \begin{bmatrix} 2 \\ -5 \\ 2 \\ 16 \end{bmatrix} \quad v_4 = \begin{bmatrix} -4 \\ 1 \\ 0 \\ -4 \end{bmatrix}$$

$$\left[\begin{array}{cccc|c} 3 & -1 & 2 & -4 & 0 \\ 2 & 3 & -5 & 1 & 0 \\ -3 & 4 & 2 & 0 & 0 \\ 4 & -12 & 16 & -4 & 0 \end{array} \right]$$

$$R_1 = R_1 / 3$$

$$\left[\begin{array}{cccc|c} 1 & -1/3 & 2/3 & -4/3 & 0 \\ 2 & 3 & -5 & 1 & 0 \\ -3 & 4 & 2 & 0 & 0 \\ 4 & -12 & 16 & -4 & 0 \end{array} \right]$$

$$R_2 = R_2 - 2R_1$$

$$\left[\begin{array}{cccc|c} 1 & -1/3 & 2/3 & -4/3 & 0 \\ 0 & 11/3 & -19/3 & 11/3 & 0 \\ -3 & 4 & 2 & 0 & 0 \\ 4 & -12 & 16 & -4 & 0 \end{array} \right]$$

$$R_3 = R_3 + 3R_1$$

$$\left[\begin{array}{cccc|c} 1 & -1/3 & 2/3 & -4/3 & 0 \\ 0 & 11/3 & -19/3 & 11/3 & 0 \\ 0 & 3 & 4 & -4 & 0 \\ 4 & -12 & 16 & -4 & 0 \end{array} \right]$$

$$R_4 = R_4 - 4R_1$$

$$\left[\begin{array}{ccc|c} 1 & -1/3 & 2/3 & -4/3 \\ 0 & 11/3 & -19/3 & 11/3 \\ 0 & 3 & 4 & -4 \\ 0 & -32/3 & 40/3 & 4/3 \end{array} \right]$$

$$R_2 = \frac{3R_2}{11}$$

$$\left[\begin{array}{ccc|c} 1 & -1/3 & 2/3 & -4/3 \\ 0 & 1 & -19/11 & 1 \\ 0 & 3 & 4 & -4 \\ 0 & -32/3 & 40/3 & 4/3 \end{array} \right]$$

$$R_1 = R_1 + \frac{R_2}{3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1/11 & -1 \\ 0 & 1 & -19/11 & 1 \\ 0 & 3 & 4 & -4 \\ 0 & -32/3 & 40/3 & 4/3 \end{array} \right]$$

$$R_3 = R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1/11 & -1 \\ 0 & 1 & -19/11 & 1 \\ 0 & 0 & 10/11 & -7 \\ 0 & -32/3 & 40/3 & 4/3 \end{array} \right]$$

$$R_4 = R_4 + \frac{32R_2}{3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1/11 & -1 \\ 0 & 1 & -19/11 & 1 \\ 0 & 0 & 101/11 & -7 \\ 0 & 0 & -56/11 & 12 \end{array} \right]$$

$$R_3 = \frac{11R_3}{101}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1/11 & -1 \\ 0 & 1 & -19/11 & 1 \\ 0 & 0 & 1 & -77/101 \\ 0 & 0 & -56/11 & 12 \end{array} \right]$$

$$R_1 = R_1 - \frac{R_3}{11}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -94/101 \\ 0 & 1 & -19/11 & 1 \\ 0 & 0 & 1 & -77/101 \\ 0 & 0 & -56/11 & 12 \end{array} \right]$$

$$R_2 = R_2 + \frac{19R_3}{11}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -94/101 \\ 0 & 1 & 0 & -32/101 \\ 0 & 0 & 1 & -77/101 \\ 0 & 0 & -56/11 & 12 \end{array} \right]$$

$$R_4 = R_4 + \frac{56}{11} R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -94/101 \\ 0 & 1 & 0 & -32/101 \\ 0 & 0 & 1 & -77/101 \\ 0 & 0 & 0 & 820/101 \end{array} \right]$$

So, the rank is 4.

$\therefore$ If the rank of the matrix of given vectors is equal to the number of vectors, the vectors are linearly independent.

Hence, the vectors are linearly independent.

$$3. A = \begin{bmatrix} 1 & 2 & 3 & 3 & 5 \\ 2 & 3 & 4 & 4 & 7 \\ 3 & 4 & 5 & 1 & 5 \\ 4 & 1 & -2 & 2 & 3 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 3 & 5 \\ 0 & -1 & -2 & -2 & -3 \\ 3 & 4 & 5 & 1 & 5 \\ 4 & 1 & -2 & 2 & 3 \end{bmatrix}$$

$$R_3 = R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 3 & 5 \\ 0 & -1 & -2 & -2 & -3 \\ 0 & -2 & -4 & -8 & -10 \\ 4 & 1 & -2 & 2 & 3 \end{bmatrix}$$

$$R_4 = R_4 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 3 & 5 \\ 0 & -1 & -2 & -2 & -3 \\ 0 & -2 & -4 & -8 & -10 \\ 0 & -7 & -14 & -10 & -17 \end{bmatrix}$$

$$R_2 = -R_2$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 3 & 5 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & -2 & -4 & -8 & -10 \\ 0 & -7 & -14 & -10 & -17 \end{array} \right]$$

$$R_1 = R_1 - 2R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -1 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & -2 & -4 & -8 & -10 \\ 0 & -7 & -14 & -10 & -17 \end{array} \right]$$

$$R_3 = R_3 + 2R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -1 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & -4 & -4 \\ 0 & -7 & -14 & -10 & -17 \end{array} \right]$$

$$R_4 = R_4 + 7R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -1 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 4 & 4 \end{array} \right]$$

$$R_3 = -\frac{R_3}{4}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -1 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 4 & 4 \end{array} \right]$$

$$R_1 = R_1 + R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 4 & 4 \end{array} \right]$$

$$R_2 = R_2 - 2R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 4 & 4 \end{array} \right]$$

$$R_4 = R_4 - 4R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x - z = 0, \quad y + 2z + v = 0, \quad w + v = 0$$

$$\begin{bmatrix} 2 \\ -2z - v \\ z \\ -v \\ v \end{bmatrix} = z \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + v \begin{bmatrix} 0 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Basis for kernel is

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

The pivot columns are one, two and fourth,

Basis for image is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \\ 2 \end{bmatrix} \right\}$$